11	Given a transaction scenario, calculate the gas fee using the gas consumed and the effective gas price.

In blockchain networks, particularly Ethereum, Gas Fee is the transaction fee paid by users to compensate validators for processing transactions and executing smart contracts.

Every operation performed on the blockchain consumes a certain amount of computational resources, measured in units called Gas.

The total gas fee depends on:

1. Gas consumed by the transaction.
2. Effective gas price per unit of gas.

Gas Fee Calculation

The gas fee is calculated using the following formula:

\[\boxed{\text{Gas Fee}=\text{Gas Consumed}\times\text{Effective Gas Price}} \]

Where:

- Gas Consumed: Actual number of gas units used to execute a transaction.
- Effective Gas Price: Actual price paid for each unit of gas, generally measured in Gwei.

Unit conversion:

$$1Gwei = 10^{-9}ETH$$

Therefore,

$$GasFee(ETH) = GasConsumed \times GasPrice(Gwei) \times 10^{-9}$$

Example

Problem:

Consider an Ethereum transaction with the following details:

- Gas consumed = 50,000 units
- Effective gas price = 30 Gwei

Calculate the total gas fee in ETH.

Step 1: Write the given values

$$GasConsumed = 50,000$$

$$EffectiveGasPrice = 30Gwei$$

Step 2: Apply the gas fee formula

$$\text{Gas Fee} = \text{Gas Consumed} \times \text{Effective Gas Price}$$

Substituting the values:

$$= 50,000 \times 30$$

$$= 1,500,000Gwei$$

Step 3: Convert Gwei into ETH

We know that:

$$1Gwei = 10^{-9}ETH$$

Therefore,

$$GasFee = 1,500,000 \times 10^{-9}$$

$$= 0.0015ETH$$

Final Answer

$$\boxed{\text{Total Gas Fee} = 0.0015\ \text{ETH}}$$

Factors Affecting Gas Fees

1. Transaction Complexity: Complex smart contract operations consume more gas than simple transactions.
2. Network Congestion: Higher network demand can increase the effective gas price.
3. Gas Consumed: More computational operations require more gas units.
4. Gas Price: Users may pay higher gas prices to get their transactions processed sooner.

4	How do liquidity pools support decentralized trading? Explain the role of liquidity providers and mention one associated risk.

Liquidity pools are collections of cryptocurrency tokens locked in smart contracts on Decentralized Exchanges (DEXs) such as Uniswap. They enable users to trade cryptocurrencies directly without requiring a traditional buyer and seller to be matched.

Instead of using an order book, decentralized exchanges use Automated Market Makers (AMMs) to facilitate trades using liquidity pools.

Liquidity Pools Support Decentralized Trading

1. A liquidity pool contains two or more types of cryptocurrency tokens, such as ETH and USDT.
2. Users deposit tokens into the pool to provide liquidity.
3. Traders exchange one token for another directly through the pool.
4. Smart contracts automatically calculate exchange rates based on token supply and demand.
5. Traders pay a small transaction fee, which is distributed among liquidity providers.

Role of Liquidity Providers (LPs)

Liquidity providers are individuals who deposit their cryptocurrency tokens into liquidity pools.

Their roles include:

- Providing liquidity: Deposit pairs of tokens (e.g., ETH and USDT).
- Enabling trading: Ensure traders can exchange tokens without waiting for another buyer or seller.
- Maintaining market efficiency: Help reduce price fluctuations caused by insufficient liquidity.
- Earning rewards: Receive a portion of transaction fees generated by trades.

In return, liquidity providers generally receive LP tokens representing their share of the pool.



Example

Suppose two liquidity providers deposit:

- 10 ETH
- 20,000 USDT

A trader wants to exchange ETH for USDT.

- The trader sends ETH to the liquidity pool.
- The smart contract automatically transfers USDT to the trader.
- The pool balances are updated.
- Liquidity providers earn a portion of the transaction fee.

Associated Risk: Impermanent Loss

Impermanent loss occurs when the price ratio of tokens in a liquidity pool changes compared with the price at which they were deposited.

For example, if the price of ETH increases significantly compared to USDT, the liquidity provider may end up with less value than if they had simply held both tokens separately.

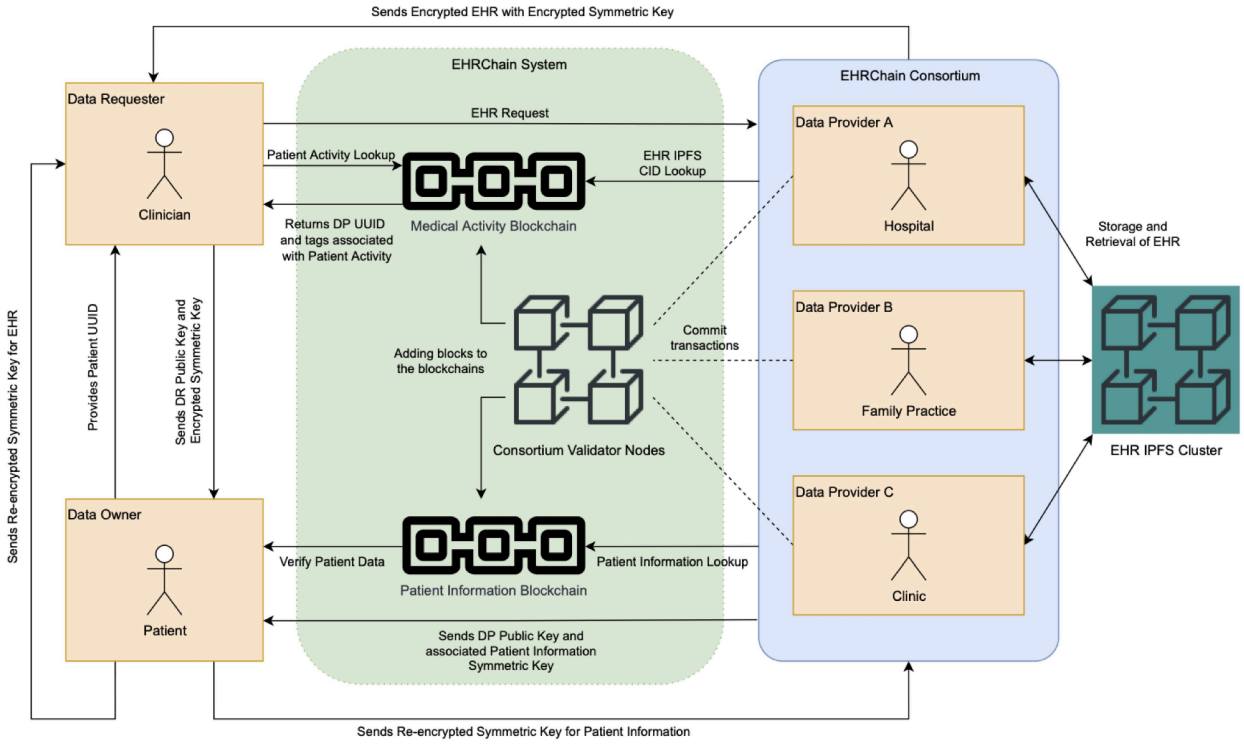
Other risks include smart contract vulnerabilities and sudden price fluctuations.

24	Demonstrate how blockchain technology can be used to provide secure and privacy-preserving sharing of sensitive healthcare records.

Blockchain technology can be used to securely share sensitive healthcare records among hospitals, doctors, laboratories, pharmacies, and patients while maintaining data privacy, integrity, transparency, and controlled access.

Traditional healthcare systems store patient records in centralized databases, which can be vulnerable to hacking, unauthorized access, data manipulation, and data breaches.

Blockchain provides a decentralized, tamper-resistant, and transparent mechanism for managing access to healthcare records without exposing sensitive patient information to unauthorized parties.



Architecture

The proposed system combines:

1. Blockchain: Maintains a tamper-resistant record of transactions, access permissions, and data-sharing activities.
2. Smart Contracts: Automatically enforce access-control policies and patient consent.
3. Off-chain Storage: Stores actual medical records in encrypted form using secure databases or IPFS.
4. Cryptography: Encrypts healthcare records to prevent unauthorized access.
5. Digital Identity: Identifies patients and authorized healthcare professionals.

architecture works as follows:

- Patient: Owns their healthcare records and controls who can access them.
- Hospital/Laboratory: Uploads medical reports in encrypted form.
- Off-chain Storage: Stores encrypted medical records securely.
- Blockchain Network: Stores the hash of medical records, access permissions, and transaction logs.
- Smart Contract: Verifies patient consent before granting access.
- Doctor: Retrieves and decrypts records only after receiving authorization.

How Privacy and Security Are Maintained

Security mechanism	Purpose
AES-256 Encryption	Protects medical records from unauthorized reading.
SHA-256 Hashing	Detects unauthorized modification of records.
Smart Contracts	Enforces patient consent and access permissions.
Digital Signatures	Verifies the identity of authorized users and transactions.
Permissioned Blockchain	Restricts participation to authorized healthcare organizations.
Access Control	Ensures only approved doctors can access specific records.
Audit Trail	Maintains a traceable history of record access.

Advantages

1. Data Privacy: Medical records remain encrypted and are accessible only to authorized users.
2. Data Integrity: Hash verification helps detect unauthorized changes.
3. Patient Control: Patients can approve or reject access requests.
4. Transparency: Every access transaction can be audited.
5. Interoperability: Different hospitals can securely exchange patient information.
6. Reduced Data Tampering: Blockchain makes recorded transactions difficult to alter without detection.

Limitations

- Blockchain implementation and maintenance can be expensive.
- Managing cryptographic keys securely is challenging.
- Healthcare organizations need compatible systems.
- Blockchain transactions may introduce latency.
- Lost private keys can create access and recovery problems.
- Revoking access cannot erase information already downloaded by an authorized user.